Unity SITL Quick-start Guide

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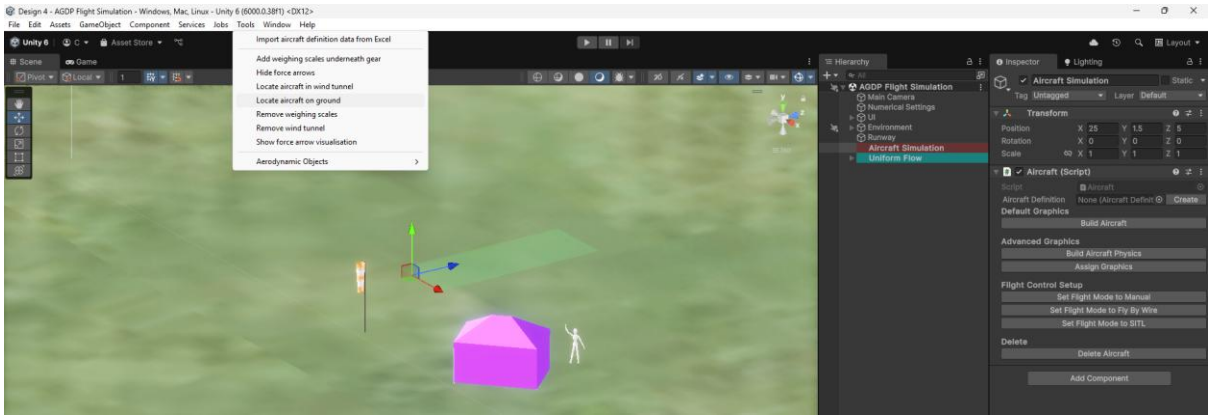
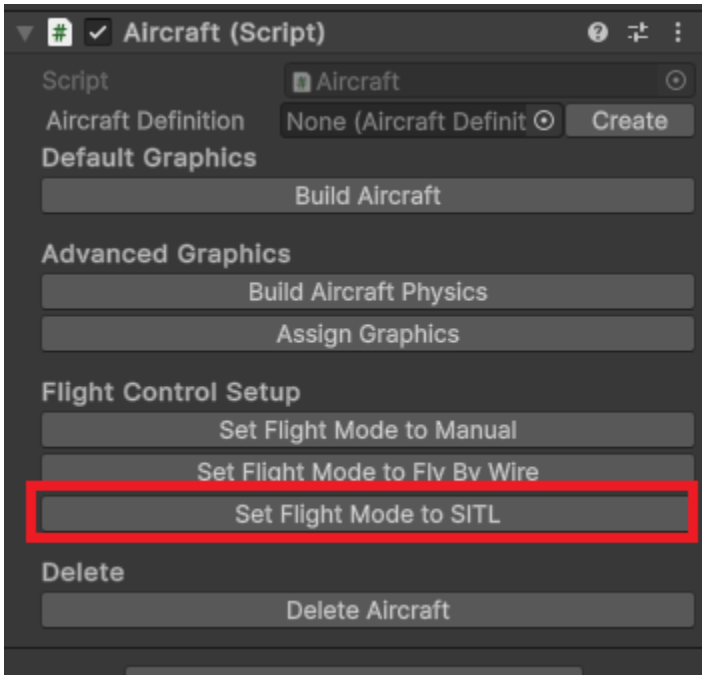
1 Unity Setup

Make sure your aircraft is located on the ground so that there are no large accelerations or velocities present when the software in the loop starts. This gives the autopilot the best chance of calibrating itself when it starts up.

Select your aircraft and go to Tools > Locate aircraft on ground

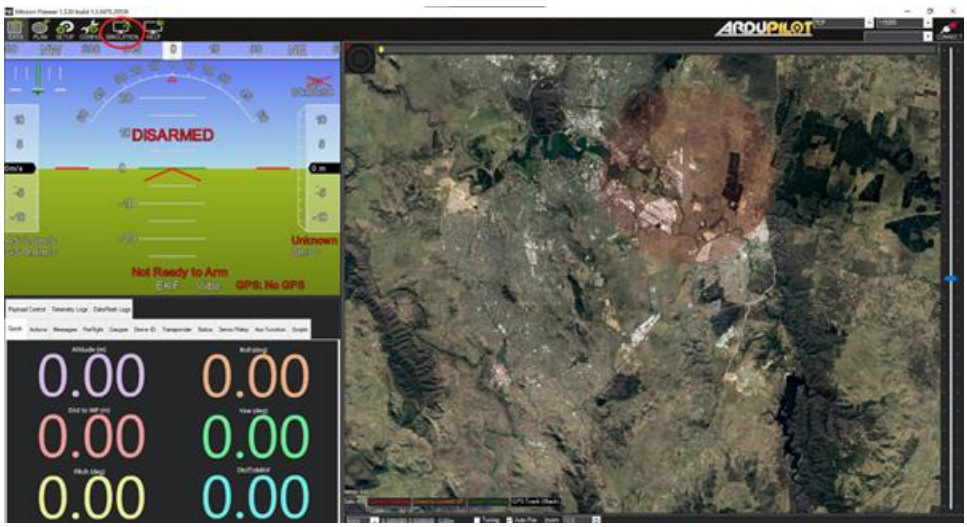


Turn on the software in the loop controls for your aircraft by clicking the “Set Flight Mode to SITL” button on your aircraft



2 Mission Planner Quick Start

Open Mission Planner and go to the Simulation tab



In the extra command line field enter:

--model json --home=53.43619,-2.250627,0,0 --defaults=C:\Users\karim\Documents\parameters.param *path to your param file*

These settings will tell mission planner about the type of simulation we want to run.

- --model json states that the physics model is external and that simulation data will be passed to mission planner using Ardupilot's JSON format
- --home sets the origin for the autopilot (lat,lon,alt,heading) as the simulation passes local NED coordinates which need to be converted into GPS lat lon alt using the home origin as a reference point.
- --defaults tells mission planner where you have saved your parameter file for your aircraft, this enables you to save changes and reload them when running a new simulation. An example of this field would be: --defaults=C:/Data/sitl.param

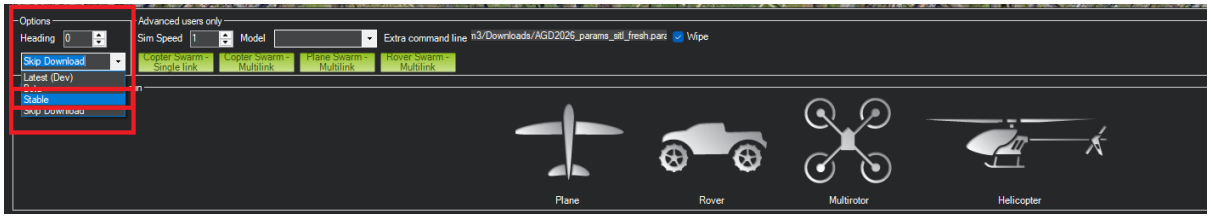
Make sure there are no spaces in your defaults path as mission planner will fail to read it properly. If you have spaces in folder names or file names, either rename them or move your file to a location where no spaces are present.

You will know if your param file failed to load because unity will fail to connect and the SITL firmware's window will display an error that looks like this:

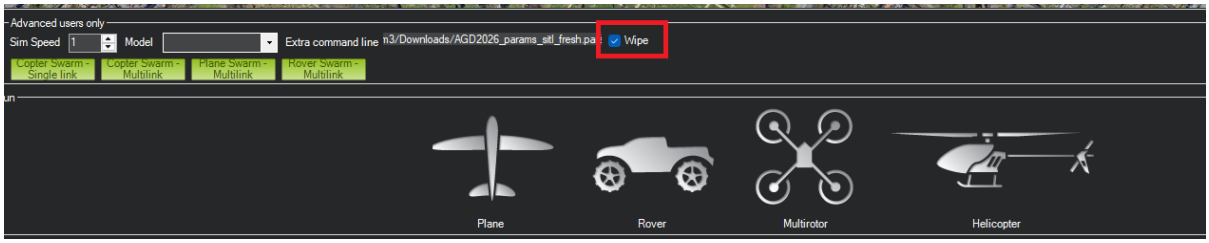
```
C:\Users\mbyx3cm3\Documents\Mission Planner\sitl\ArduPlane.exe
Setting SIM_SPEEDUP=1.000000
Starting SITL: JSON
Home: 53.436190 -2.250627 alt=80.000000m hdg=0.000000
JSON control interface set to 127.0.0.1:9002
Starting sketch 'ArduPlane'
Starting SITL input
Using Irlock at port : 9005
bind port 5760 for SERIAL0
SERIAL0 on TCP port 5760
PANIC: Failed to load defaults from C:/Users/mbyx3cm3/Downloads/AGD2026_params

Running: sh dumpstack.sh 134 >dumpstack.sh_ArduPlane.134.out 2>&1
Failed
Running: sh dumpcore.sh 134 >dumpcore.sh_ArduPlane.134.out 2>&1
Failed
```

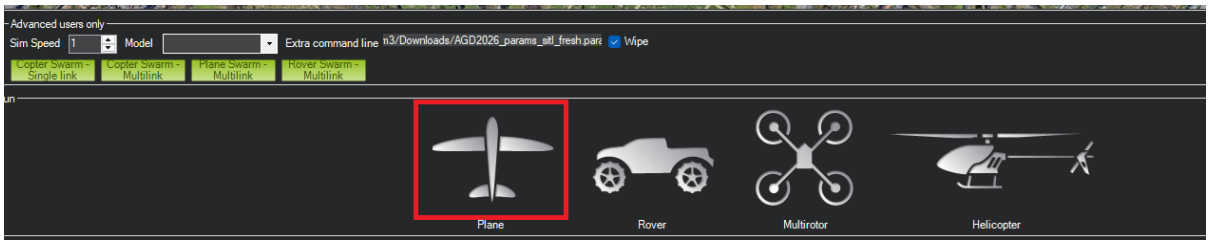
For the first run you will need to download the ArduPlane firmware. This is done by selecting “Stable” in the options dropdown menu



It may also help to check the “wipe” box to clear any session values from mission planner



Finally, to start the SITL firmware click the Plane icon



On your first run through, Mission Planner will download the SITL software. After it has downloaded it, you can select “skip download” in the options for future simulations.

When Mission Planner starts to connect, start your unity scene by pressing play in unity.

3 Troubleshooting and tips

Make sure you have nothing selected in the Unity hierarchy window as this will slow your simulation down.

If the unity simulation is laggy, try clicking the *pause* button twice. This refreshes the communication order between unity and ArduPilot and helps with laggy instances

Look out for the **SAFE** message in mission planner. This means that your safety switch is engaged and the autopilot will send no signals to anything on the aircraft. You can disable this by changing the BRD_SAFETY_DEFLT parameter. **Make sure you only do this on a SITL copy of your parameter file – this must be kept on for your test flights**